

MOULIKA MANTHRIPRAGADA

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PROFESSIONAL SUMMARY

DevOps Engineer with 4 years of experience designing and automating cloud infrastructure on Amazon Web Services. Strong expertise in Terraform-based Infrastructure as Code, CI/CD pipeline engineering using Jenkins and Git, and container orchestration with Docker and Amazon EKS. Hands-on experience deploying scalable applications leveraging EC2, VPC, RDS, Auto Scaling, and Application Load Balancers. Proven ability to enhance system reliability through blue-green deployments, DevSecOps integration, and observability using Prometheus, Grafana, ELK Stack, and AWS CloudWatch. Skilled in improving release velocity, strengthening cloud security, and supporting high-availability production environments aligned with SRE practices.

TECHNICAL SKILLS

Cloud Platform: Amazon Web Services (AWS): EC2, EKS, ECS, S3, VPC, IAM, RDS, DynamoDB, Route 53, CloudFront, ALB/NLB, Auto Scaling, CloudWatch, CloudTrail, SNS, SQS, Systems Manager, Secrets Manager

CI/CD & Release Engineering: Git, Jenkins, GitHub Actions, GitLab CI/CD, AWS CodePipeline, AWS CodeBuild, AWS CodeDeploy, Pipeline as Code, Deployment Automation

Containerization & Kubernetes: Docker, Kubernetes, Amazon EKS, Helm, Kustomize, Ingress Controllers, Horizontal Pod Autoscaling, RBAC

Infrastructure as Code: Terraform, AWS CloudFormation, Infrastructure Provisioning, Multi-Environment Automation

Configuration & Automation: Ansible, Configuration Management, Application Deployment Automation

Monitoring & Observability: Prometheus, Grafana, ELK Stack, AWS CloudWatch, Log Aggregation, Alerting

DevSecOps & Cloud Security: SonarQube, Trivy, Snyk, OWASP Dependency Check, Container Security, IAM Governance, Secrets Management

Scripting: Bash, Python, Groovy, Shell, YAML, JSON, REST APIs

Build & Artifact Management: Maven, Gradle, Nexus, JFrog Artifactory

Systems & Reliability: Linux Administration, VPC Networking, DNS, TCP/IP, Load Balancing, SSL/TLS, Incident Management, Root Cause Analysis, High Availability

PROFESSIONAL EXPERIENCE

DevOps Engineer

Apr 2021 - Oct 2023

Wipro Technologies | Chennai, India

- Spearheaded the design of CI/CD pipelines using Jenkins, Git, and Maven, embedding automated testing and artifact management to cut release turnaround time by 30% and stabilize deployments across AWS environments.
- Built and maintained AWS infrastructure through Terraform provisioning of EC2 Auto Scaling, VPC networking, RDS Multi-AZ, and ALB, reducing manual setup effort by 70% while strengthening production availability.
- Standardized multi-environment infrastructure by creating reusable Terraform modules, accelerating environment provisioning by 40% and ensuring configuration consistency across development, QA, and production accounts.
- Migrated monolithic workloads to Docker containers and orchestrated deployments on Amazon EKS with rolling updates and Horizontal Pod Autoscaling, improving peak-time scalability by 35% for retail traffic spikes.
- Orchestrated blue-green and canary releases through Kubernetes-integrated Jenkins pipelines, enabling near zero downtime deployments and improving release confidence during high-volume sales events.
- Established centralized observability using Prometheus, Grafana, and AWS CloudWatch to monitor application and infrastructure metrics, reducing Mean Time to Resolution by 40% through proactive alerting.
- Automated Linux server configuration and middleware deployments with Ansible playbooks, eliminating configuration drift and increasing deployment consistency across distributed AWS systems.
- Embedded DevSecOps practices by integrating SonarQube code analysis and Trivy image scanning into CI/CD workflows, lowering critical vulnerabilities in production builds by 25% and enhancing cloud security compliance.

Cloud Engineer

Oct 2019 - Mar 2021

Wipro Technologies | Chennai, India

- Built AWS-based application environments using EC2, S3, VPC, and Application Load Balancers to support retail workloads, reducing environment setup time by 35% and improving deployment consistency.
- Automated multi-environment infrastructure provisioning with Terraform modules, eliminating configuration inconsistencies and lowering manual setup errors by 40% across QA and staging accounts.
- Integrated Jenkins pipelines with Git repositories to validate build artifacts and deployment packages, improving release success rates to 95% through structured CI/CD validation.
- Deployed Dockerized services into Amazon EKS clusters for pre-production validation, accelerating container verification cycles by 30% during sprint-based releases.
- Established monitoring dashboards using AWS CloudWatch and Grafana to track compute metrics and application logs, decreasing incident response time by 25% through real-time alerting.
- Strengthened access control by configuring IAM roles, security groups, and subnet isolation within VPC architecture, enhancing cloud security posture across non-production environments.
- Validated deployment workflows and rollback mechanisms within Jenkins-driven releases, reducing post-deployment defects and improving release stability for customer-facing features.

- Streamlined Linux instance configuration and middleware setup using Ansible automation, minimizing configuration drift and ensuring consistent application behavior across AWS-hosted systems.

PROJECTS

Enterprise CI/CD & Kubernetes Platform Implementation

- Engineered a Jenkins-driven CI/CD framework integrated with Git, Maven, Docker, AWS CodePipeline, and Terraform to automate build, test, and Amazon EKS deployments, reducing release time from 45 minutes to under 10 minutes for high-traffic retail applications.
- Deployed containerized microservices onto Amazon EKS using Helm charts, Kubernetes Ingress Controllers, RBAC, and Horizontal Pod Autoscaling, sustaining peak seasonal traffic with zero downtime across Auto Scaling-backed EC2 worker nodes.
- Implemented Prometheus and Grafana monitoring integrated with AWS CloudWatch logs and metrics to track pod health, infrastructure utilization, and application performance, lowering Mean Time to Resolution by 40% through real-time alerting.

Python Microservices Library Platform

- Developed and containerized Python-based microservices using Docker with REST APIs for modular service communication, improving deployment efficiency by 60% through standardized container builds and environment isolation.
- Provisioned supporting infrastructure using Terraform on AWS including EC2, VPC networking, and Application Load Balancers to enable scalable service hosting and controlled network segmentation.
- Configured Jenkins pipelines integrated with Git to automate Docker image builds and deployment validation, while leveraging ELK Stack and CloudWatch for centralized logging and runtime visibility across distributed services.

DevSecOps Pipeline & Container Security Implementation

- Integrated SonarQube static analysis into Jenkins CI pipelines connected to Git repositories, improving code quality governance and reducing critical vulnerabilities in release builds by 30%.
- Implemented Docker image security scanning using Trivy and Snyk within CI workflows, blocking non-compliant container artifacts from deployment into Amazon EKS environments.
- Consolidated pipeline and access audit logs through ELK Stack ingestion with AWS CloudWatch log streaming, improving security monitoring and traceability across CI/CD operations.

EDUCATION

Master of Science in Computer & Information Sciences

Concordia University Wisconsin | USA

Oct 2023 - Dec 2025

Bachelor of Technology in Electronics & Communication Engineering

Aurora's Technological & Research Institute | Hyderabad

Aug 2016 - May 2020

CERTIFICATIONS

- Solutions Architecture Job Simulation - **AWS**
- Cloud DevOps - **Intel Corporation**
- AWS Fundamentals Specialization - **Coursera**
- Continuous Delivery & DevOps - **Coursera**
- Learning Jenkins - **LinkedIn Learning**
- Docker Foundations Professional Certificate - **LinkedIn Learning**